



Beyond Keywords: Building Spatial AI for Local Map Intelligence

A technical deep-dive into hybrid local search architecture — solving the "near me" failure for campus and urban map platforms.

LOCAL SEARCH · SPATIAL AI · NLP



The "Near Me" Failure in Current Map Engines

Local Intent Failure

A query for **"quiet places near me"** in Pilani returns results 1,000+ km away in Bengaluru or Hyderabad — because the engine prioritizes keyword matches over spatial proximity.

Semantic Blindness

Systems match literal names like **"Quiet Lands Gate"** rather than interpreting "quiet" as a qualitative characteristic — low noise, calm environment, low crowd density.

📄 **Consequence:** Rapid trust erosion. Users abandon platforms for competitors when core local functionality fails.

Architecture Exploration: Rejected Options

DESIGN DECISIONS

Option 1 — Frontier Model Per Request (GPT-4 / Groq)

Verdict: Rejected. High marginal inference costs per query and significant hallucination risk around database schema. Not economically viable at scale for a local-first product.

Option 2 — Custom In-House Spatial LLM from Scratch

Verdict: Rejected for Phase 1. Requires months of development, massive GPU clusters, and a large labeled spatial dataset. Capital-intensive, slow to execute, and premature for an MVP.

Both options were evaluated against cost, latency, hallucination risk, and time-to-deploy. Neither met the bar for Phase 1 viability.

Recommended Architecture: The Hybrid Logic

LOCAL SQL ENGINE + CLOUD CONVERSATIONAL WRAPPER



Intent Engine

Uses **defog/llama-3-sqlcoder-8b** hosted locally to translate natural language into strict PostGIS queries — no ambiguity, no schema drift.



Deterministic Core

AI is restricted to **intent extraction only**. PostGIS execution and parameter binding are fully deterministic — guaranteeing **zero SQL hallucinations**.



Cloud Wrapper

Outputs are routed to the **Groq API** for high-speed conversational formatting — including Hinglish responses tailored for the campus user base.



The "Local RAG" Transformation

Replacing the external cloud API with a fully grounded, offline-capable local SLM pipeline.

1

Goal

100% offline inference with **zero marginal cost** per query at any scale.

2

Method

A fine-tuned **Small Language Model** (e.g., Llama-3.2 1B) paired with a local RAG retrieval pipeline.

3

Grounding

The SLM retrieves campus-specific knowledge and live SQL results to generate accurate, context-aware responses — no cloud dependency.

SLM Prompt & Context Strategy

TEACHING THE AI TO HANDLE POSTGIS JSON OUTPUT

Input → Output Pipeline

Input: User question + deterministic JSON output from PostGIS.

Constraint: The SLM is instructed to ignore raw geometry metadata and focus solely on `name`, `amenity`, and `vibe` fields.

Output style: Conversational, local-language aware, and spatially grounded.

Live Example

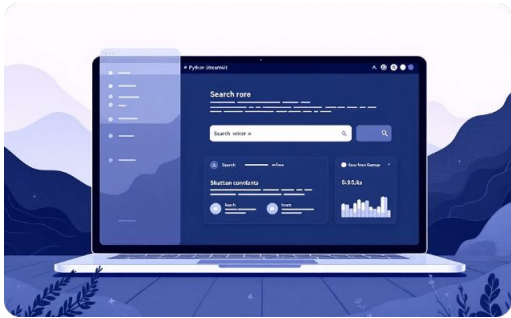
Query: "Study spot near me?"

PostGIS JSON: `[{'name': 'Library'}]`

SLM Response: "Bhai, Library chala ja, wahan focus accha milega."

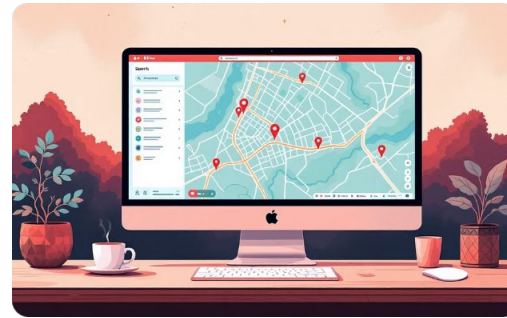
- ❏ The SLM never sees raw geometry. It only reasons over semantic, human-relevant attributes.

UI & Product Evolution



Phase 1 — Streamlit Prototype

Rapid prototyping of the conversational search bar and intent preview. Enabled fast feedback loops and early validation of the query pipeline.



Phase 2 — Production Web App

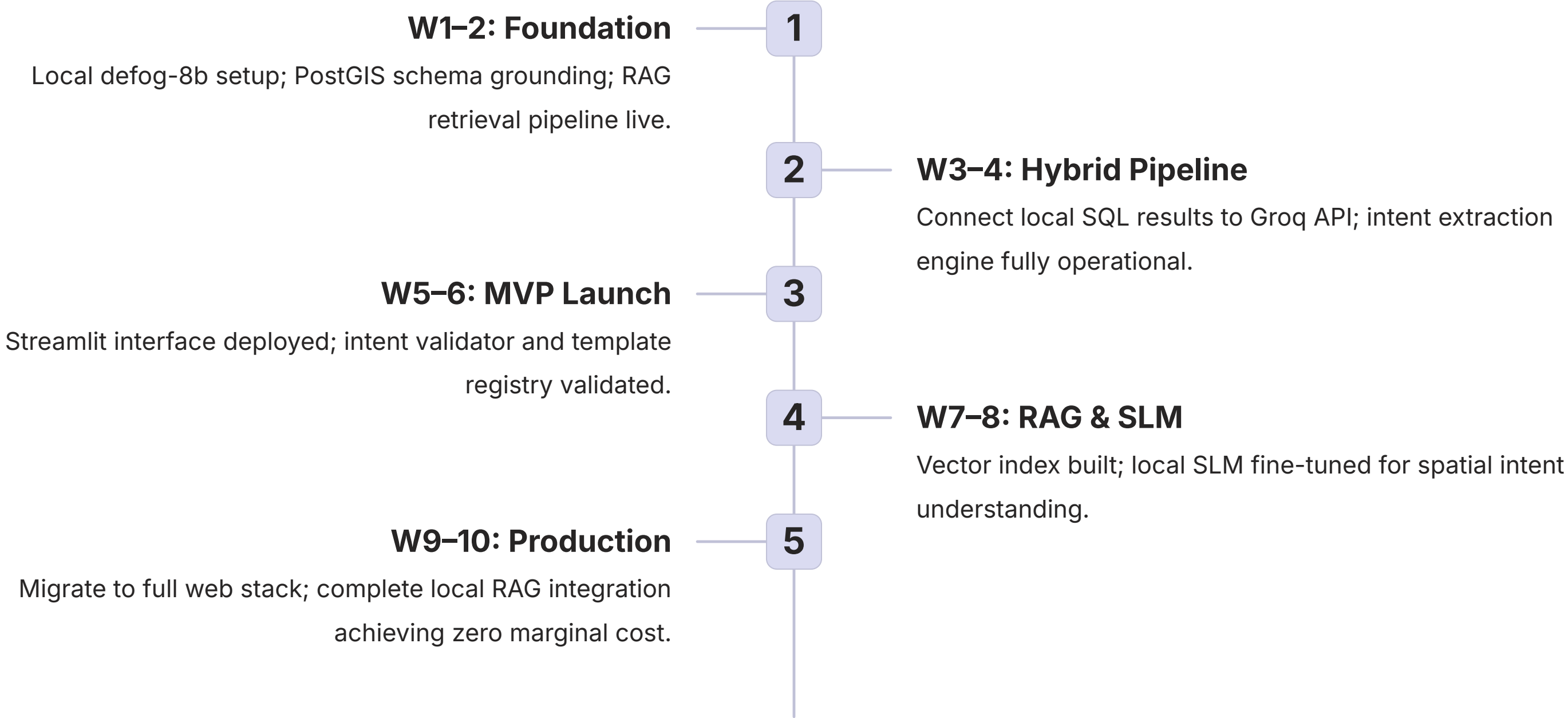
Deployed via **FastAPI** (backend) and **React/Next.js** (frontend) to support concurrent users, production-grade stability, and future feature extensibility.



Feature Integrations

Real-time **confidence indicators** and smart nearby ranking based on semantic relevance combined with spatial proximity scores.

10-Week Implementation Roadmap



Transition to Local SLM (W5-6)

CLOUD-TO-EDGE AI

Building on the hybrid pipeline established in W3-4, our focus in W5-6 shifts to achieving complete local inference, eliminating external cloud dependencies.



Phase 1: Groq API Dependency

Leveraged **Groq API** for high-speed conversational formatting, routing PostGIS outputs to the cloud.

Maintained conversational quality but incurred per-query costs and cloud dependency.



Phase 2: Local RAG + Llama

Deployed a fine-tuned **Llama model** with a local RAG retrieval pipeline, enabling 100% offline inference.

Achieved zero marginal cost while maintaining the high quality of conversational responses.

Strategic Positioning

Building the next generation of map intelligence — moving well beyond keyword matching.

Beyond Keywords

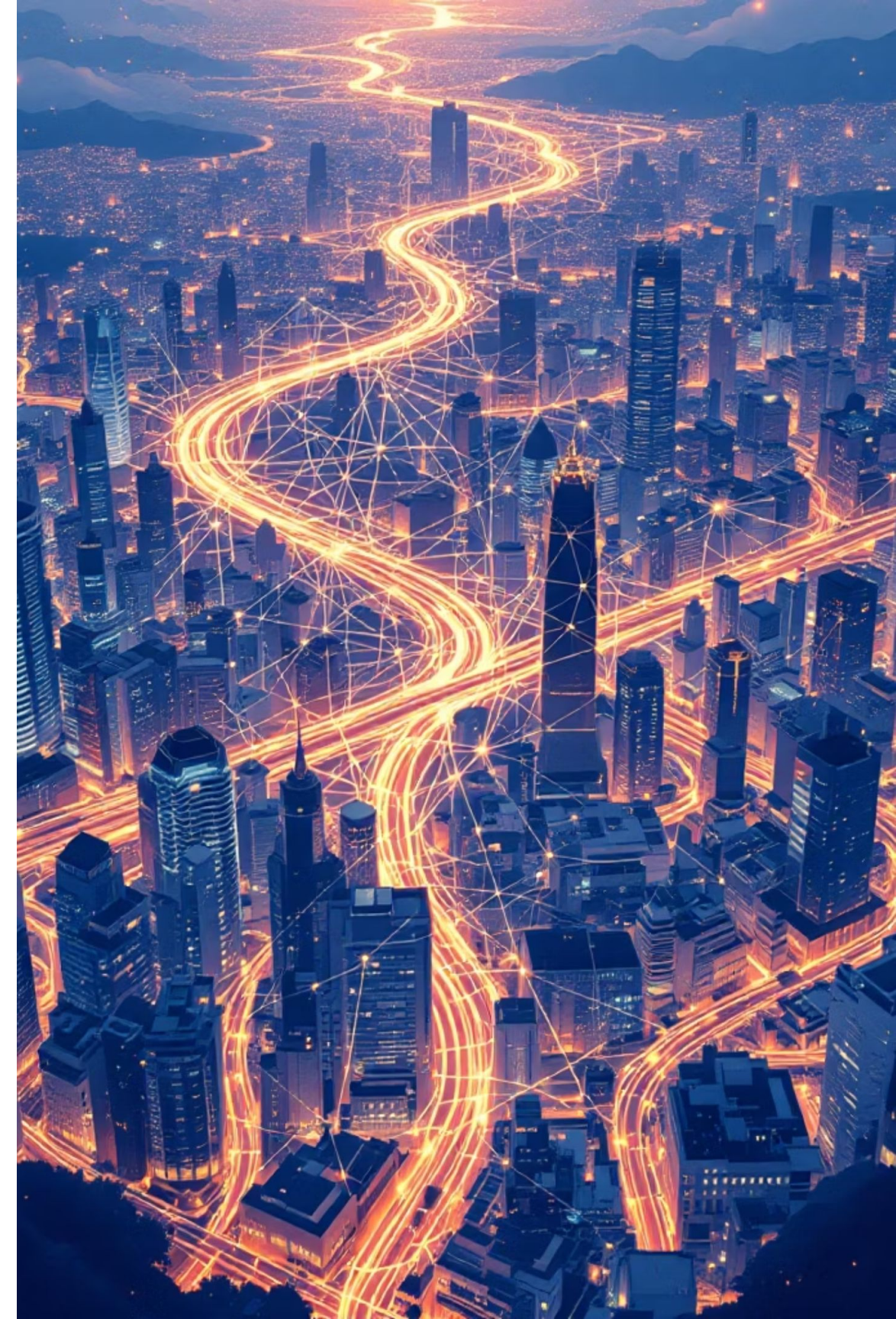
Shift from literal word matching to intelligent discovery based on **experiential needs** — quiet, crowd-level, ambiance, and context.

Research-Backed Ranking

Top-K frequent spatial keyword algorithms deliver **smarter, more relevant** location rankings with measurable precision gains.

Zero-Cost Scaling

Local inference ensures the system stays **economically sustainable** as the user base grows — no per-query cloud bills.



Key Takeaways & Next Steps

Hybrid

Architecture

Balances
hallucination risk
and inference cost
(W3-4)

Semantic

Spatial AI

Replaces brittle
keyword matching
with experiential
understanding



Local RAG + Llama

Eliminates cloud
dependency with
100% offline
inference (W5-6
onwards)

Immediate Next Steps

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1

Finalize defog-8b schema grounding

Lock PostGIS schema and test intent extraction
precision on real query logs.

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2

Launch Streamlit MVP (W5-6)

Deploy to internal stakeholders for structured
feedback before production migration.

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3

Begin SLM fine-tuning dataset

Curate labeled spatial query examples for
Llama-3.2 1B fine-tuning in W7-8.

Future Expansion & Innovation



GROWTH OPPORTUNITIES

Our strategic roadmap includes key initiatives to expand capabilities, enhance user experience, and secure a long-term competitive advantage in spatial AI.



Expand POI Coverage

Broaden our spatial intelligence beyond campus to include local restaurants, shops, and essential services, unlocking new markets and user segments.



Advanced Spatial UI

Introduce intuitive map-based filtering for proximity, amenity types, and real-time crowd density, empowering users with visual, interactive exploration.



Proprietary Spatial SLM

Develop a custom Small Language Model, fine-tuned specifically for complex spatial queries, delivering unparalleled accuracy and a significant competitive edge.